

A.1 Kinematics

Past Paper Questions

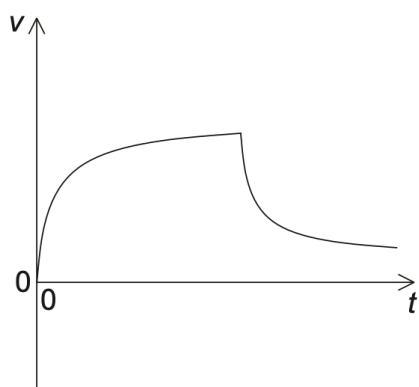
These are past paper questions selected from past papers (2016 - 2022) that relate to the topic A.1 Kinematics.

Answers can be found in the separate accompanying document

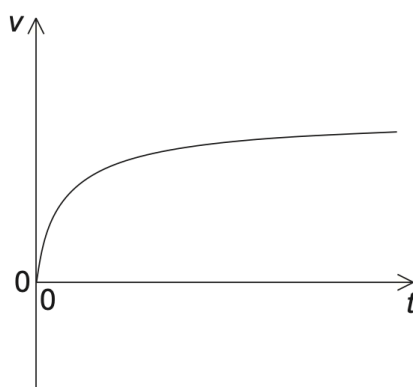
Question 1

An aircraft is moving horizontally. A parachutist leaves the aircraft and a few seconds later opens her parachute. Which graph shows the variation of the vertical speed v with time t for the parachutist from the time she leaves the aircraft until just before landing?

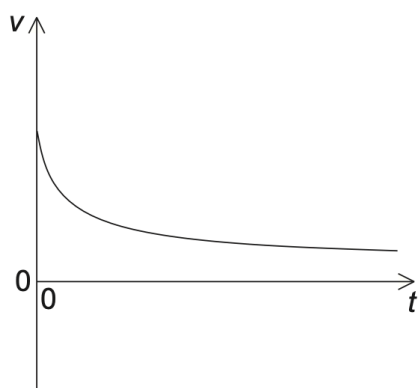
A.



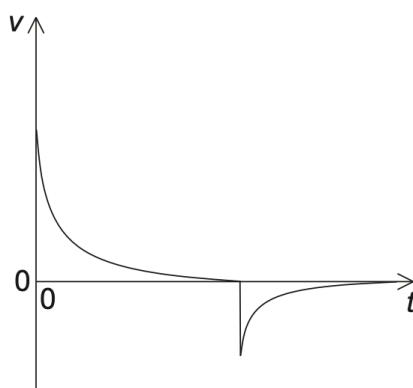
B.



C.



D.



- **Learning Objective:** A.1.9
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2016, Paper 1
- **Question type:** Multiple Choice

Question 2

A car moves north at a constant speed of 3 m s^{-1} for 20 s and then east at a constant speed of 4 m s^{-1} for 20 s. What is the average speed of the car during this motion?

- A. 7.0 m s^{-1}
- B. 5.0 m s^{-1}
- C. 3.5 m s^{-1}
- D. 2.5 m s^{-1}

- **Learning Objective:** A.1.3
- **Level:** Standard Level and Higher Level
- **Past Paper:** Nov 2016, Paper 1
- **Question type:** Multiple Choice

Question 3

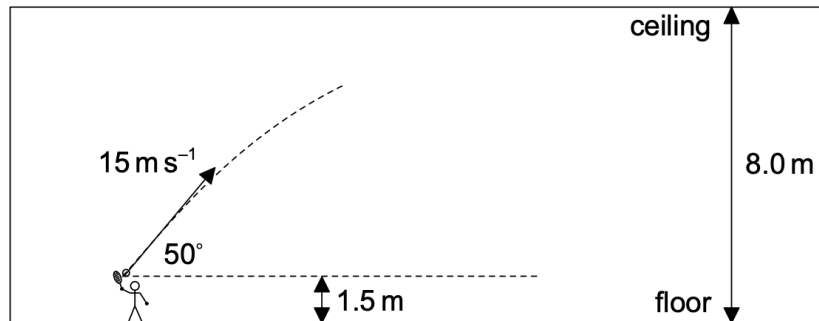
An object of weight W is falling vertically at a constant speed in a fluid. What is the magnitude of the drag force acting on the object?

- A. 0
- B. $\frac{W}{2}$
- C. W
- D. $2W$

- **Learning Objective:** A.1.9
- **Level:** Standard Level and Higher Level
- **Past Paper:** Nov 2016, Paper 1
- **Question type:** Multiple Choice

Question 4

A tennis ball is hit with a racket from a point 1.5 m above the floor. The ceiling is 8.0 m above the floor. The initial velocity of the ball is 15 m s^{-1} at 50° above the horizontal. Assume that air resistance is negligible.



(a) Determine whether the ball will hit the ceiling.

[3]

- **Learning Objective:** A.1.7, A.1.8
- **Level:** Standard Level and Higher Level
- **Past Paper:** Nov 2016, Paper 2
- **Question type:** Calculate

Question 5

Which of the following is a scalar quantity?

- A. Velocity
- B. Momentum
- C. Kinetic energy
- D. Acceleration

- **Learning Objective:** A.1 Math Skill 1
- **Level:** Standard Level
- **Past Paper:** May 2017, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 6

An object is released from rest in the gravitational field of the Earth. Air resistance is negligible. How far does the object move during the fourth second of its motion?

- A. 15m
- B. 25m
- C. 35m
- D. 45m

- **Learning Objective:** A.1.6
- **Level:** Standard Level
- **Past Paper:** May 2017, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 7

Which is a vector quantity?

- A. Pressure
- B. Electric current
- C. Temperature
- D. Magnetic field

- **Learning Objective:** A.1 Math Skill 1
- **Level:** Standard Level
- **Past Paper:** May 2017, Paper 1, TZ2
- **Question type:** Multiple Choice

Question 8

A ball is tossed vertically upwards with a speed of 5.0 m s^{-1} . After how many seconds will the ball return to its initial position?

- A. 0.50 s
- B. 1.0 s
- C. 1.5 s
- D. 2.0 s

- **Learning Objective:** A.1.6
- **Level:** Standard Level
- **Past Paper:** May 2017, Paper 1, TZ2
- **Question type:** Multiple Choice

Question 9

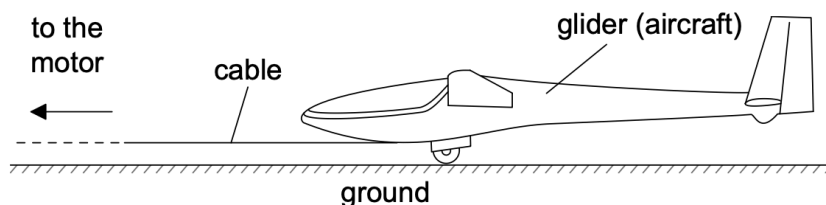
A projectile is fired horizontally from the top of a cliff. The projectile hits the ground 4 s later at a distance of 2 km from the base of the cliff. What is the height of the cliff?

- A. 40 m
- B. 80 m
- C. 120 m
- D. 160 m

- **Learning Objective:** A.1.8
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2017, Paper 1, TZ2
- **Question type:** Multiple Choice

Question 10

A glider is an aircraft with no engine. To be launched, a glider is uniformly accelerated from rest by a cable pulled by a motor that exerts a horizontal force on the glider throughout the launch.



- (a) The glider reaches its launch speed of 27.0 m s^{-1} after accelerating for 11.0 s . Assume that the glider moves horizontally until it leaves the ground. Calculate the total distance travelled by the glider before it leaves the ground.

[2]

- **Learning Objective:** A.1.6
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2017, Paper 2, TZ2
- **Question type:** Calculate

Question 11

An object is released from a stationary hot air balloon at height h above the ground. An identical object is released at height h above the ground from another balloon that is rising at constant speed. Air resistance is negligible. What does **not** increase for the object released from the rising balloon?

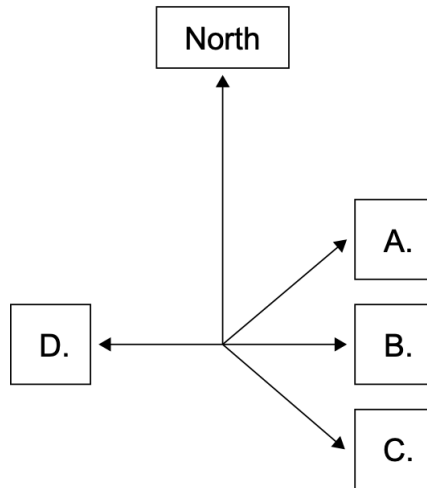
- A. The distance through which it falls
- B. The time taken for it to reach the ground
- C. The speed with which it reaches the ground
- D. Its acceleration

- **Learning Objective:** A.1.5
- **Level:** Standard Level and Higher Level
- **Past Paper:** Nov 2017, Paper 1
- **Question type:** Multiple Choice

Question 12

A river flows north. A boat crosses the river so that it only moves in the direction east of its starting point.

What is the direction in which the boat must be steered?



- **Learning Objective:** A.1 Math Skill 3
- **Level:** Standard Level
- **Past Paper:** May 2018, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 13

An object is projected vertically upwards at time $t = 0$. Air resistance is negligible. The object passes the same point above its starting position at times 2 s and 8 s.

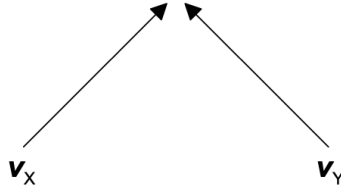
If $g = 10 \text{ m s}^{-2}$, what is the initial speed of the object?

- A. 50
- B. 30
- C. 25
- D. 4

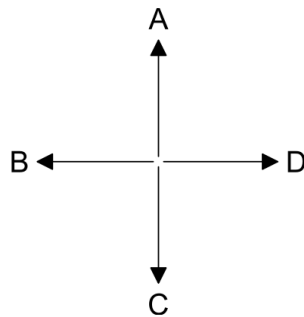
- **Learning Objective:** A.1.6
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2018, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 14

The velocities $\mathbf{v}_X$ and $\mathbf{v}_Y$ of two boats, X and Y, are shown.



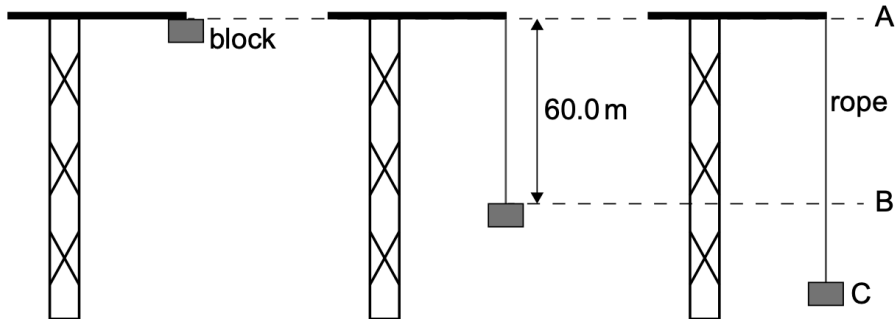
Which arrow represents the direction of the vector $\mathbf{v}_X - \mathbf{v}_Y$?



- **Learning Objective:** A.1 Math Skill 3
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2018, Paper 1, TZ2
- **Question type:** Multiple Choice

Question 15

An elastic climbing rope is tested by fixing one end of the rope to the top of a crane. The other end of the rope is connected to a block which is initially at position A. The block is released from rest. The mass of the rope is negligible.



The unextended length of the rope is 60.0 m. From position A to position B, the block falls freely.

(a) At position B the rope starts to extend. Calculate the speed of the block at position B. [2]

- **Learning Objective:** A.1.6
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2018, Paper 2, TZ1
- **Question type:** Calculate

Question 16

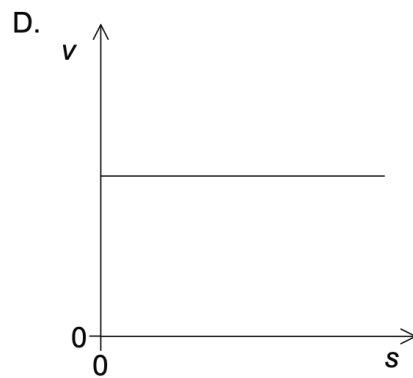
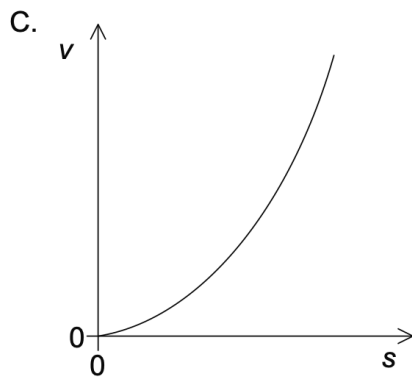
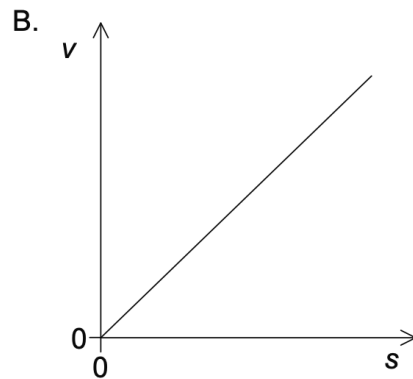
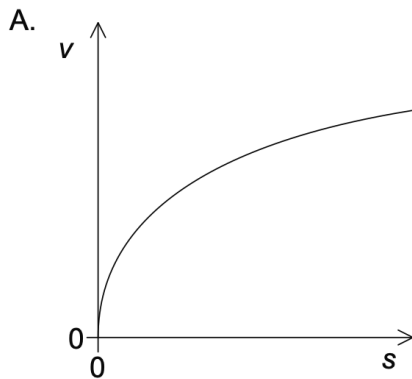
A truck has an initial speed of 20 m s^{-1} . It decelerates at 4.0 m s^{-2} . What is the distance taken by the truck to stop?

- A. 2.5 m
- B. 5.0 m
- C. 50 m
- D. 100 m

- **Learning Objective:** A.1.6
- **Level:** Standard Level and Higher Level
- **Past Paper:** Nov 2018, Paper 1
- **Question type:** Multiple Choice

Question 17

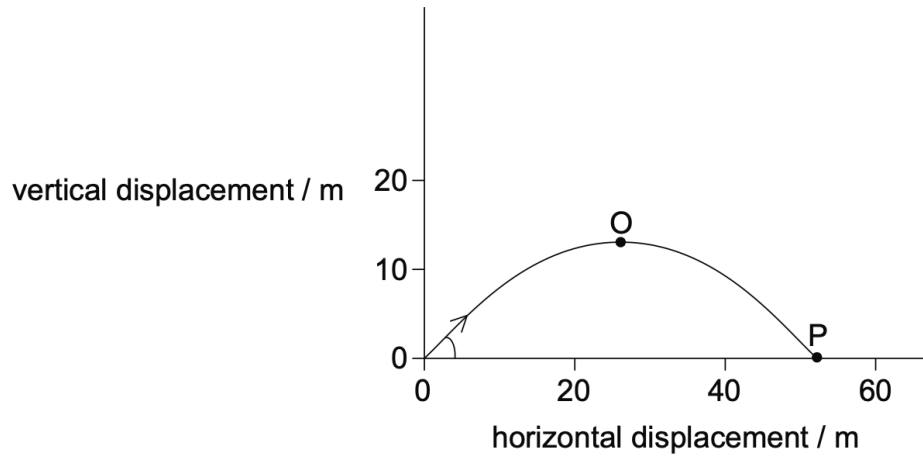
A runner starts from rest and accelerates at a constant rate throughout a race. Which graph shows the variation of speed v of the runner with distance travelled s ?



- **Learning Objective:** A.1.6
- **Level:** Standard Level and Higher Level
- **Past Paper:** Nov 2018, Paper 1
- **Question type:** Multiple Choice

Question 18

A projectile is fired at an angle to the horizontal. Air resistance is negligible. The path of the projectile is shown.



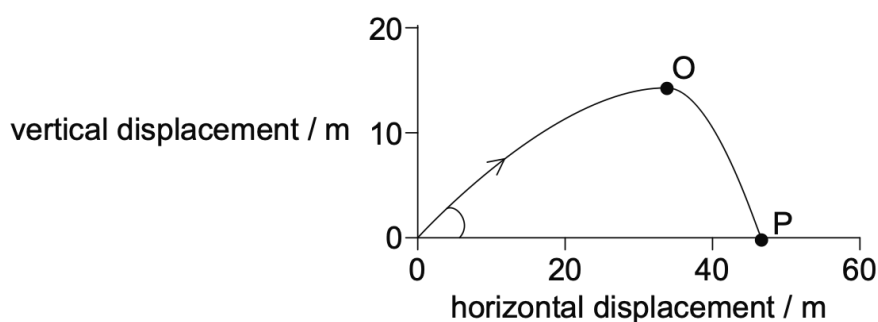
Which gives the magnitude of the horizontal component and the magnitude of the vertical component of the velocity of the projectile between O and P?

	Magnitude of horizontal component of velocity	Magnitude of vertical component of velocity
A.	constant	increases
B.	constant	constant
C.	increases	increases
D.	increases	constant

- **Learning Objective:** A.1.7
- **Level:** Standard Level
- **Past Paper:** Nov 2018, Paper 1
- **Question type:** Multiple Choice

Question 19

A projectile is fired at an angle to the horizontal. The path of the projectile is shown.



Which gives the magnitude of the horizontal component and the magnitude of the vertical component of the velocity of the projectile between O and P?

	Magnitude of horizontal component of velocity	Magnitude of vertical component of velocity
A.	decreases	increases
B.	decreases	constant
C.	constant	increases
D.	constant	constant

- **Learning Objective:** A.1.7 and A.1.9
- **Level:** Higher Level
- **Past Paper:** Nov 2018, Paper 1
- **Question type:** Multiple Choice

Question 20

A sky diver is falling at terminal speed when she opens her parachute. What are the direction of her velocity vector and the direction of her acceleration vector before she reaches the new terminal speed?

	Direction of velocity vector	Direction of acceleration vector
A.	upwards	upwards
B.	upwards	downwards
C.	downwards	upwards
D.	downwards	downwards

- **Learning Objective:** A.1.6 and A.1 Math Skill 6
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2019, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 21

A stone is thrown downwards from the edge of a cliff with a speed of 5.0 ms^{-1} . It hits the ground 2.0 s later. What is the height of the cliff?

- A. 20m
- B. 30m
- C. 40m
- D. 50m

- **Learning Objective:** A.1.6
- **Level:** Standard Level
- **Past Paper:** May 2019, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 22

A ball is thrown upwards at an angle to the horizontal. Air resistance is negligible. Which statement about the motion of the ball is correct?

- A. The acceleration of the ball changes during its flight.
- B. The velocity of the ball changes during its flight.
- C. The acceleration of the ball is zero at the highest point.
- D. The velocity of the ball is zero at the highest point.

- **Learning Objective:** A.1.8
- **Level:** Standard Level
- **Past Paper:** May 2019, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 23

A sports car is accelerated from 0 to 100 km per hour in 3 s. What is the acceleration of the car?

- A. $0.1g$
- B. $0.3g$
- C. $0.9g$
- D. $3g$

- **Learning Objective:** A.1.6
- **Level:** Higher Level
- **Past Paper:** May 2019, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 24

A girl throws an object horizontally at time $t = 0$. Air resistance can be ignored. At $t = 0.50$ s the object travels horizontally a distance x in metres while it falls vertically through a distance y in metres.

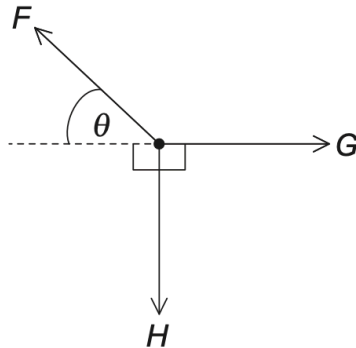
What is the initial velocity of the object and the vertical distance fallen at $t = 1.0$ s?

	Initial velocity / m s^{-1}	Vertical distance fallen / m
A.	x	$2y$
B.	x	$4y$
C.	$2x$	$2y$
D.	$2x$	$4y$

- **Learning Objective:** A.1.8
- **Level:** Higher Level
- **Past Paper:** May 2019, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 25

An object is held in equilibrium by three forces of magnitude F , G and H that act at a point in the same plane.



Three equations for these forces are

- I. $F \cos \theta = G$
- II. $F = G \cos \theta + H \sin \theta$
- III. $F = G + H$

Which equations are correct?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

- **Learning Objective:** A.1 Math Skill 5
- **Level:** Standard Level
- **Past Paper:** Nov 2019, Paper 1
- **Question type:** Multiple Choice

Question 26

A ball falls from rest in the absence of air resistance. The position of the centre of the ball is determined at one-second intervals from the instant at which it is released. What are the distances, in metres, travelled by the centre of the ball during **each** second for the first 4.0 s of the motion?

- A. 5, 10, 15, 20
- B. 5, 15, 25, 35
- C. 5, 20, 45, 80
- D. 5, 25, 70, 150

- **Learning Objective:** A.1.6
- **Level:** Higher Level
- **Past Paper:** Nov 2019, Paper 1
- **Question type:** Multiple Choice

Question 27

An object is thrown from a cliff at an angle to the horizontal. The ground below the cliff is horizontal.

Three quantities are known about this motion.

- I. The horizontal component of the initial velocity of the object
- II. The initial angle between the velocity of the object and the horizontal
- III. The height of the cliff

What are the quantities that must be known in order to determine the horizontal distance from the point of projection to the point at which the object hits the ground?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

- **Learning Objective:** A.1.8
- **Level:** Higher Level
- **Past Paper:** Nov 2019, Paper 1
- **Question type:** Multiple Choice

Question 28

A list of four physical quantities is

- acceleration
- energy
- mass
- temperature

How many scalar quantities are in this list?

- A. 1
- B. 2
- C. 3
- D. 4

- **Learning Objective:** A.1 Math Skill 1
- **Level:** Standard Level
- **Past Paper:** Nov 2020, Paper 1
- **Question type:** Multiple Choice

Question 29

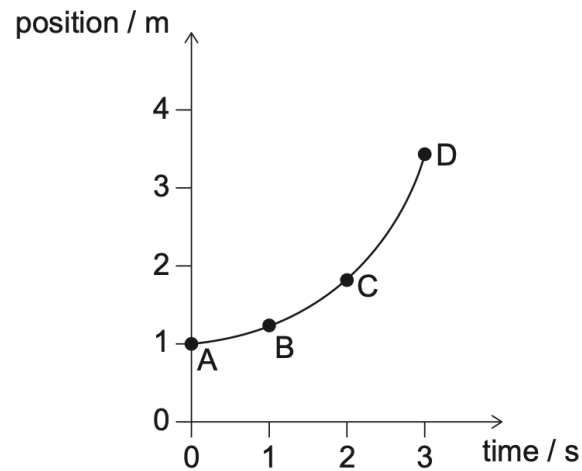
A large stone is dropped from a tall building. What is correct about the speed of the stone after 1 s?

- A. It is decreasing at increasing rate.
- B. It is decreasing at decreasing rate.
- C. It is increasing at increasing rate.
- D. It is increasing at decreasing rate.

- **Learning Objective:** A.1.6
- **Level:** Standard Level
- **Past Paper:** May 2021, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 30

The graph shows how the position of an object varies with time in the interval from 0 to 3 s.



At which point does the instantaneous speed of the object equal its average speed over the interval from 0 to 3 s?

- **Learning Objective:** A.1.3
- **Level:** Standard Level
- **Past Paper:** May 2021, Paper 1, TZ1
- **Question type:** Multiple Choice